Enhancing Your User Experience

The Systematic Destruction of Professional Software Expertise Through Interface Redesign Cycles

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Abstract

Professional software interfaces change on release cycles that have no correlation with changes to the underlying functional capabilities. The phrase "enhancing your user experience" signals the replacement of a user's accumulated operational expertise with a new interface that serves the production metrics of a UI design team. This paper documents the pattern across seven major professional tools spanning 27 years, quantifies the retraining cost imposed on users, identifies the organizational incentive structure that drives the pattern, and demonstrates that the functional kernels of these applications have been stable for 10-25 years while their interfaces have been reorganized 3-8 times per decade. The paper concludes that the only structural defense against this pattern is to make the interface a user-owned data layer rather than a designer-shipped artifact.

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1. The Inversion

The phrase "enhancing your user experience" contains a precise inversion. It refers to the replacement of one experience with another. The experience being replaced is the one the user built: the spatial memory that locates a button in 200 milliseconds, the keyboard shortcut chain that executes a five-step operation in under two seconds, the procedural workflow that moves from intent to output without conscious navigation. This accumulated experience is the user's primary asset when operating the software. It is the difference between a novice and an expert. It is the reason an expert produces more output per hour than a novice using the same tool.

The enhancement destroys this asset. The user's spatial memory is invalidated when panels move. The shortcut chain breaks when keystrokes are reassigned. The procedural workflow fails when dialogs are inserted, merged, or split. The expert is converted into a novice for a relearning period that ranges from days to weeks depending on the scope of the change.

The new experience that replaces the old one is not requested by the user. It is produced by a UI design team operating on a release cycle. The design team's output is the change itself. The change is the product. The user's relearning cost is the price.

Two definitions of "user experience" exist simultaneously within the organizations that produce professional software:

User experience (actual): The accumulated operational knowledge a user holds about how to accomplish their tasks within the software. This knowledge is stored in muscle memory, spatial memory, and procedural memory. It is built through thousands of hours of repetition. It is the reason experienced users are faster than new users. It is destroyed by interface changes and rebuilt through forced repetition with the new interface.

User experience (artifact): The designed interface that a UI team produces. It exists as wireframes, design system documentation, component libraries, and interaction specifications. It is measured by design system compliance scores, visual consistency audits, accessibility checklists, and first-time-user comprehension tests. It is the thing that appears in release notes and marketing materials.

These two definitions are in direct conflict. Optimizing the artifact requires changing the interface. Changing the interface destroys the actual experience. The organization measures the artifact. The user bears the cost of the destroyed experience. The cost is invisible to the organization because it is distributed across millions of users and does not appear in any metric the organization tracks.

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2. The Stable Kernel

Before examining what changes, it is necessary to establish what does not change. The functional kernel of each major professional tool has been stable for a decade or more. The operations that users perform have not changed. The underlying algorithms, data structures, and processing pipelines have not changed in ways that require interface reorganization.

Product	Core Operations	Stable Since	Years Stable
Adobe Photoshop	Layers, selections, brushes, masks, transforms, color adjustments, filters	2003 (CS)	23

Product	Core Operations	Stable Since	Years Stable
Autodesk AutoCAD	Lines, arcs, circles, polylines, dimensions, hatches, blocks, layers, plot	2006	20
Unreal Engine	Actors, components, blueprints, materials, sequencer, build	2014 (UE4)	12
Unity	GameObjects, components, inspector, hierarchy, build	2018 (2018 LTS)	8
Blender	Modeling, sculpting, texturing, animation, rendering, compositing	2019 (2.80)	7
DaVinci Resolve	Media management, editing, color grading, mixing, delivery	2015 (Fusion integration)	11
Figma	Frames, components, auto layout, prototyping, export	2019	7

Genuine functional additions do occur. Photoshop added Generative Fill in 2023. Unreal Engine added Nanite and Lumen in 2022. Blender added Geometry Nodes in 2021. DaVinci Resolve added neural engine tools in 2023. These are real new capabilities that justify new interface elements because there is a new operation the user needs to access.

The additions are rare. Between Photoshop CS (2003) and Generative Fill (2023), twenty years passed with no functional change that required any interface reorganization. During those twenty years, the interface was reorganized approximately eight times. Each reorganization was described as an enhancement. None was required by a change in what the software does.

The functional kernel is the proof. If the operations the user performs have not changed, then any change to how the user accesses those operations is not a functional requirement. It is an organizational output.

3. The Task Audit

For each exemplar product, the following tables track the most common user operations across release versions. Each entry records the interaction path: the sequence of keystrokes, clicks, and visual scans required to complete the operation. Changes in interaction count that are not caused by changes in functionality are interface reorganization events.

3.1 Adobe Photoshop

Task	1999 (5.5)	2012 (CS6)	2015 (CC 2015)	2024 (CC 2024)
Export to PNG	`Alt+Shift+Ctrl+S` → PNG → Save (3 steps)	Same (3 steps)	File → Export → Export As → PNG → configure → Save (6 steps)	Same as 2015, or assign Quick Export shortcut manually (6 steps default)
Flatten and save	`Ctrl+Shift+S` → `Ctrl+S` (2 steps)	Same (2 steps)	Flatten → Save As → format dialog → options → confirm (5 steps)	Flatten → Save As vs Save a Copy disambiguation → format → options → confirm (5-6 steps)
Resize image	`Ctrl+Alt+I` → type width → OK (3 steps)	Same (3 steps)	Same (3 steps)	Same (3 steps)
New layer	`Ctrl+Shift+N` → Enter (2 steps)	Same (2 steps)	Same (2 steps)	Same (2 steps)
Adjust levels	`Ctrl+L` → adjust → OK (3 steps)	Same, Properties panel added (3 steps + visual noise)	Same (3 steps)	Same (3 steps)
Crop	`C` → drag → Enter (3 steps)	Same (3 steps, options bar reorganized)	Same (3 steps, options bar reorganized again)	Same (3 steps, options bar reorganized again)

Task	1999 (5.5)	2012 (CS6)	2015 (CC 2015)	2024 (CC 2024)
Refine selection edge	Select → Refine Edge → adjust → OK (4 steps)	Same (4 steps)	Select → Select and Mask → new workspace → adjust → OK (5 steps, new layout)	Same as 2015 (5 steps)
Apply Gaussian Blur	Filter → Blur → Gaussian Blur → adjust → OK (5 steps)	Same (5 steps)	Same (5 steps)	Same (5 steps)
Content-Aware Fill	Edit → Fill → Content-Aware → OK (4 steps)	Same (4 steps)	Same (4 steps)	Edit → Content-Aware Fill → new workspace → adjust → OK (5 steps)
Batch process	File → Automate → Batch → configure → OK (5 steps)	Same (5 steps)	Same (5 steps)	Same (5 steps)

The operations that gained steps (Export, Flatten+Save, Refine Edge, Content-Aware Fill) are operations where the UI team introduced a new dialog, workspace, or panel. The operations that remained unchanged (Resize, New Layer, Levels, Crop, Gaussian Blur, Batch) are operations where the UI team did not introduce a new interaction surface.

The functional capability behind each operation is identical across all versions. A PNG export in 1999 and a PNG export in 2024 produce the same output. The difference is the number of interactions required to reach that output.

3.2 Autodesk AutoCAD

Task	2000	2009 (Ribbon)	2020	2026
Draw line	`L` → click → click → Enter (4 steps)	Same (4 steps)	Same (4 steps)	Same (4 steps)
Draw circle	`C` → click → drag → click (4 steps)	Same (4 steps)	Same (4 steps)	Same (4 steps)

Task	2000	2009 (Ribbon)	2020	2026
Print drawing	`Ctrl+P` → configure → OK (3 steps, one dialog)	`Ctrl+P` → expand sections → configure → OK (4-5 steps, collapsible sections)	`Ctrl+P` → expand sections → configure → OK (reorganized again) (4-5 steps)	Same (4-5 steps)
Set current layer	Layer dropdown → select (2 steps)	Layer dropdown → select (2 steps, widget redesigned)	Same (2 steps, widget redesigned again)	Same (2 steps)
Insert block	`I` → select → OK → place (4 steps)	Same (4 steps)	Same (4 steps)	Same (4 steps)
Add dimension	`DIM` → click → click → place (4 steps)	Same (4 steps)	Same (4 steps)	Same (4 steps)
Export to PDF	Print → PDF printer → OK (3 steps)	Same (3 steps) or `EXPORTPDF` (typed, no shortcut)	Same (3 steps)	Same (3 steps)
Hatch area	`H` → click boundary → Enter (3 steps, dialog-based)	`H` → ribbon panel appears → click → Enter (3 steps, ribbon-based, layout changed)	Same (3 steps, ribbon reorganized)	Same (3 steps)
Edit polyline	`PE` → select → options → Enter (4 steps)	Same (4 steps)	Same (4 steps)	Same (4 steps)
Toggle ORTHO	F8 (1 step)	F8 (1 step, status bar icon changed from text to icon)	F8 (1 step, icon redesigned)	F8 (1 step)

The command-line operations (`L`, `C`, `I`, `DIM`, `PE`, `H`) are completely unchanged because the command line is a text interface. The UI team cannot redesign text. The GUI operations (Print, Layer dropdown, Hatch ribbon panel, status bar) were reorganized because they are visual surfaces the UI team can modify.

The command line is the existence proof that interface stability is possible. AutoCAD's command line has been stable for 44 years. The commands work the same way in 2026 as they did in 1982. The Ribbon, introduced in 2009 as a "modern" replacement for toolbars and menus, has been reorganized four times in 17 years.

3.3 Unreal Engine

Task	UE4 (2014)	UE5.0 (2022)	UE5.4 (2026)
Place actor	Content Browser → drag to viewport (2 steps)	Content Browser → drag to viewport (2 steps, browser redesigned)	Same (2 steps)
Build lighting	Toolbar → Build → Build Lighting Only (2 clicks)	Menu → Build → Build Lighting Only (button removed from toolbar) (3 clicks)	Menu → Build → submenu → Build Lighting Only (4 clicks)
Create Blueprint	Content Browser → right-click → Blueprint (3 clicks)	Same (3 clicks, browser redesigned)	Same (3 clicks)
Edit Blueprint variable	Details → variable row → edit (2 clicks)	Details → variable row → edit (2 clicks, panel reorganized)	Same (2 clicks)
Create material	Content Browser → right-click → Material (3 clicks)	Same (3 clicks)	Same (3 clicks)
Package project	File → Package → platform (3 clicks)	Platforms → platform → Package (3 clicks, menu restructured)	Same (3 clicks)
Create button widget	Palette → drag Button → set style (3 steps + minimal config)	Palette → drag Button → configure Normal/Hovered/Pressed/Disabled/Focused styles (3 steps + 5-9 state configs)	Same (3 steps + 9 state configs)
Play in editor	Toolbar → Play (1 click)	Toolbar → Play (1 click, button moved slightly)	Same (1 click)

The button widget row is examined in detail in Section 5.

3.4 DaVinci Resolve

Task	Resolve 14 (2017)	Resolve 17 (2021)	Resolve 19 (2024)
Color grade clip	Color page → select → wheels (2 steps)	Same (2 steps)	Same (2 steps)
Cut clip	`Ctrl+B` (1 step)	Same (1 step)	Same (1 step)
Add transition	Drag from panel (2 steps)	Same (2 steps)	Same (2 steps)
Export timeline	Deliver → format → Add → Start (4 steps)	Deliver → format → Add → Start (4 steps, page reorganized)	Deliver → format → Add → Start (4 steps, page reorganized again) or Quick Export (no default shortcut)
Import media	Media page → drag files (2 steps)	Same (2 steps, page reorganized)	Same (2 steps, page reorganized again)
Add audio effect	Fairlight → select track → effects rack (3 steps)	Same (3 steps, rack moved)	Same (3 steps, rack moved again)

The color grading operations have zero interaction changes across 7 years. The color page is Resolve's competitive advantage. Colorists are the users who choose Resolve over competitors. The UI team does not touch the color page because doing so would cause measurable user attrition. Every other page has been reorganized. This selectivity proves the UI team makes strategic decisions about which users can absorb retraining cost and which users would leave.

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4. The Export Problem

The final operation in every creative workflow is getting output out of the application. The user has finished their work. They need to produce a deliverable file. This operation sits between "work is done" and "work is delivered." Every additional interaction on this operation is pure friction between completion and delivery.

Across every major creative application, the export operation has become slower over time.

4.1 Photoshop: Export to PNG

1999: `Alt+Shift+Ctrl+S` opened Save for Web. The dialog showed a preview and format options. The user selected PNG, clicked Save, chose a location, and confirmed. The shortcut was four keys held simultaneously, then two clicks. The dialog remembered the last-used format. A user who always exported PNG opened the dialog with the format already set. Total: 1 shortcut + 1 click + 1 file dialog.

2015: Save for Web was deprecated. The label "(Legacy)" was appended to the menu entry. The promoted replacement was Export As, accessed through File → Export → Export As. Export As opens a larger dialog with format selection, canvas size controls, metadata toggle, color space options, and a preview that re-renders on every settings change. The dialog does not remember the last-used format across sessions in all configurations. No default keyboard shortcut was assigned to Export As.

The user who exports to PNG 50 times a day went from a four-key shortcut to a three-level menu navigation. The menu path is: move mouse to File (menu bar), click, move to Export (submenu), hover to expand, move to Export As, click. This is 5 pointer movements and 2 clicks before the dialog opens. The dialog then requires format selection (1 click), and Save (1 click + file dialog). Total: 7 interactions minimum, versus 3 with the shortcut.

2017: Quick Export as PNG was added under File → Export → Quick Export as PNG. This performs a direct PNG export with no dialog, using the last-used settings. No default keyboard shortcut was assigned. The user can assign one through Edit → Keyboard Shortcuts → File → Export → Quick Export as PNG. The assignment path is: Edit (menu bar) → Keyboard Shortcuts → scroll or search to find "Quick Export as PNG" → click the shortcut field → press the desired key combination → click Accept → click OK. This is approximately 8 interactions to configure a shortcut that should have been default.

2024: The state is unchanged. Quick Export exists. It has no default shortcut. A fresh Photoshop installation on a new machine has no fast path to PNG export. The user must either configure Quick Export's shortcut (8 interactions, repeated per machine) or use Export As (7+ interactions per export).

The progression: a three-interaction operation in 1999 became a seven-interaction operation in 2024, with an optional configuration step of eight interactions that must be repeated on every new installation to restore the original speed.

4.2 Illustrator: Export to SVG

2005 (CS2): File → Save As → SVG. The SVG Options dialog had four settings: SVG Profile, Fonts, Images, and CSS Properties. The dialog was one panel. The user set options once and the dialog remembered them.

2024: File → Export → Export As → SVG. The SVG Options dialog now has: Styling (Presentation Attributes / Internal CSS / Inline Style), Font (SVG / Outline / Converted), Images (Embed / Link), Object IDs (Layer Names / Minimal / Unique), Decimal (1-7), Minify, Responsive, and additional output options. Twelve configurable fields. The dialog does not consistently remember settings across sessions. The user who wants the same SVG output every time must verify twelve fields every time.

4.3 Premiere Pro: Export Video

2017 (CC 2017): File → Export → Media opened the Export Media dialog. One panel with Source, Output, and codec settings. Format dropdown, preset dropdown, output path. Configure and click Export.

2023 (CC 2023): The Export Media dialog was replaced with a full-page Export workspace. The workspace has a source monitor, output preview, format/preset selectors, encoding settings, and a render button distributed across multiple panels. The workspace layout is different from the previous dialog. Users who knew the dialog layout had to relearn positions. The workspace offers more simultaneous visibility of settings but requires more screen space and more visual scanning to locate specific options.

The operation is: "export this timeline as H.264 MP4." The functional requirement has not changed. The interface surface area approximately doubled.

4.4 Blender: Render and Save

2.79 (2018): `F12` rendered the current frame. In the render window, `F3` saved the result. Two keystrokes for the complete render-and-save workflow.

2.80 (2019): `F12` still renders. `F3` was reassigned to the operator search menu. Saving the render result requires: in the render window, Image menu → Save As → file dialog. The two-keystroke workflow became a keystroke plus three interactions.

4.x (2026): Same state. `F3` remains assigned to operator search. No replacement shortcut was assigned for saving the render result. The user can assign one through Edit → Preferences → Keymap → Image Editor → Image → Save As Image → assign shortcut. This configuration path is approximately 8 interactions deep.

4.5 The Pattern

Product	Operation	Original Steps	Current Steps	Steps Added
Photoshop	Export PNG	3	7 (or 3 after 8-step config)	+4 (or +8 once)
Illustrator	Export SVG	5 (4 memorable options)	7 (12 options to verify)	+2 steps, +8 cognitive
Premiere Pro	Export MP4	5 (one dialog)	7 (full workspace)	+2
Blender	Render + Save	2	5 (or 2 after 8-step config)	+3 (or +8 once)
Resolve	Export timeline	4	4 (page reorganized) or Quick Export (no shortcut)	+0 steps, +2 relearning events
AutoCAD	Print to PDF	3	5 (expanded sections)	+2
Figma	Export frame	2 (panel visible)	4 (panel collapsed, must expand)	+2

No product made its export operation faster across any major release. Every product either added steps, removed a shortcut, or reorganized the interface for the export operation. The operation that the user performs most frequently at the end of every work session — the operation where speed has the highest impact on workflow — is the operation that has consistently become slower.

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5. The Button Problem

Unreal Engine's UMG widget system provides the most concentrated example of design-system complexity imposed on users. A button is the most fundamental interactive element in any user interface. The user's requirement is: a rectangular region that responds to a click by executing a function, with visual feedback that it can be clicked.

5.1 What the User Wants

A button requires two pieces of information to function: what it looks like and what it does when clicked. In the simplest case, the user provides a background color, a text label, and a click handler. The button should display the label on the background, change appearance slightly when the mouse hovers over it, change appearance when the mouse presses it, and execute the handler when the mouse releases on it.

5.2 What UE Requires

UE's Slate/UMG button widget exposes the following states, each requiring independent visual configuration:

State	Description
Normal	Default appearance, no interaction
Hovered	Mouse cursor is over the button
Pressed	Mouse button is held down on the button
Disabled	Button is non-interactive
Focused	Button has keyboard focus
Normal + Selected	Button is in selected state, no hover
Hovered + Selected	Button is in selected state, mouse hovering
Pressed + Selected	Button is in selected state, mouse pressing
Disabled + Selected	Button is in selected state, non-interactive

Each state requires a style definition containing: background brush (which may be a solid color, a texture, a material instance, or a nine-slice image), tint color (RGBA), outline color, outline width, corner radius, padding (four edges independently), margin (four edges independently), minimum desired size, maximum desired size, foreground color, and optionally a sound event.

5.3 The Configuration Burden

For a button with textured backgrounds (the standard in shipped game UI), each of the nine states requires an image asset. A button bar with five buttons requires 45 image assets. A menu screen with twenty buttons across multiple groups requires 180 image assets. A full game UI with eight menu screens, each containing 15-25 buttons, requires 1,080 to 1,800 image assets for buttons alone, before considering other widget types (checkboxes, sliders, tabs, text inputs).

For a button with solid-color backgrounds (the standard in tools and prototypes), each state requires manual color specification. There is no auto-derivation system that generates hover (lighter), pressed (darker), and disabled (desaturated) states from a base color. The user must manually specify each color for each state. Five buttons with solid-color backgrounds require 45 manual color entries.

5.4 The Design System Justification

The justification for this configuration surface is accessibility and polish. A shipped game UI should respond to all interaction states with distinct visual feedback. Keyboard focus indication is an accessibility requirement. Disabled state indication prevents user confusion. Selected state indication provides context in toggle button groups.

This justification is valid for the final polish pass of a shipped AAA game's player-facing UI. It is not valid for:

- Prototyping (where buttons need to work, not look finished)
- Editor tools (where developers build internal tools with minimal UI investment)
- Game jams (where the entire game is built in 48 hours)
- Early development (where gameplay mechanics are tested through placeholder UI)
- Any project where the team does not include a dedicated UI artist

The system does not offer a graduated path. There is no "simple button" that requires two configuration values (background color, text color) and auto-derives the remaining states. The user must configure the full state matrix or accept the default style, which is a flat gray rectangle that does not match any project's visual identity.

5.5 The Specialization Requirement

The complexity of the button configuration system means that configuring a game's UI requires specialized knowledge. A gameplay programmer who can implement combat systems, AI behavior trees, and networked replication cannot set up a menu screen without spending hours learning the UMG widget style system.

Studios respond by hiring dedicated UI artists and UI programmers. These are roles whose entire purpose is configuring widget styles, creating UI materials, and building UMG widget blueprints. The roles exist because the tool's complexity requires them. The tool's complexity is maintained because the roles exist to justify it. The UI team

within Epic that designs the UMG system builds for the complexity requirements of the UI artists at AAA studios who provide feedback. The feedback loop reinforces the complexity because the people providing feedback are the people whose jobs depend on the complexity.

5.6 The Simple Alternative

A system that solves the same problem with minimal configuration:

The user specifies two values: background color and text color. The system derives:

State	Derivation
Normal	Background color as specified
Hovered	Background color lightened 10%
Pressed	Background color darkened 10%
Disabled	Background color at 50% saturation, 70% opacity
Focused	Normal + 2px outline in text color
Selected	Background color shifted 15% toward a highlight hue
Hovered + Selected	Selected color lightened 10%
Pressed + Selected	Selected color darkened 10%
Disabled + Selected	Selected color at 50% saturation, 70% opacity

Any state can be individually overridden. The system provides the simple case as the default and the complex case as an opt-in override. This approach covers approximately 90% of use cases with two configuration values. The remaining 10% of cases override individual states as needed.

UE does not provide this. Every state must be explicitly configured. The 90% case pays the cost of the 10% case on every button.

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6. The Incentive Machine

The pattern documented in Sections 2-5 is not accidental. It is the output of an organizational incentive structure that requires interface changes on a release cycle.

6.1 The UI Team's Production Metric

A UI design team inside a software company must produce measurable output. The output must be visible. Performance improvements are credited to engineering. Bug fixes are credited to QA. Backend architecture improvements are credited to platform engineering. The only artifacts that are visibly and exclusively the UI team's work are changes to the interface the user sees.

If the UI team ships a release with no visible changes, the team produced no measurable output. A team with no measurable output is reduced in the next budget cycle. The UI

team's continued existence requires visible interface changes on every release cycle.

This is not a failure of individual designers. It is a structural incentive. The designers may be talented, thoughtful, and genuinely concerned with user welfare. The organizational structure requires them to produce changes regardless of whether changes are needed. A designer who says "the current interface is optimal, I recommend no changes" is a designer who produced no output for the release cycle.

6.2 The Measurement Gap

The metrics used to evaluate UI changes do not capture the cost of the changes.

Metrics that are measured:

Metric	What It Captures	Who Benefits
First-time user task completion rate	Whether new users can find operations	New users
Visual consistency score	Whether the interface matches the design system	Design team
Accessibility compliance	Whether the interface meets WCAG guidelines	Users with disabilities
User survey (post-release)	Immediate reaction to new interface	Marketing (positive quotes)
A/B test (new users)	Whether new layout improves new user performance	New users

Metrics that are not measured:

Metric	What It Would Capture	Who Bears the Cost
Power user velocity change	Whether experienced users are faster or slower	Power users
Shortcut retention rate	Whether existing shortcuts still work	All existing users
Relearning time	Hours spent rebuilding spatial and procedural memory	All existing users
Custom configuration loss	Whether user-configured workspaces, shortcuts, or presets survived the update	Power users
Documentation update cost	Hours spent updating internal training materials	Teams and organizations
Error rate during relearning	Whether users make more mistakes during the transition period	All existing users

The metrics that justify the change measure the new interface's performance with users who have no prior experience. The metrics that would quantify the cost measure the change's impact on users who had prior experience. The first set is measured. The second set is not. The decision to ship the change is made using only the first set.

6.3 The Subscription Coupling

Subscription licensing models remove the user's ability to remain on a version that works for them.

Adobe Creative Cloud requires the current version. An organization paying for Creative Cloud receives interface changes with every update. There is no option to pay for Creative Cloud and receive only bug fixes and performance improvements without interface reorganization. The functional features (Generative Fill, Neural Filters) are bundled with the interface changes. The user who wants the new features must accept the new interface.

Autodesk subscriptions follow the same model. AutoCAD subscribers receive the current version with the current ribbon layout. There is no "keep my current ribbon but give me the new DWG features" option.

Unreal Engine and Unity are free to use but couple interface changes with engine updates. A studio that needs a bug fix in the rendering pipeline must update to a version that includes interface reorganization. There is no patch-only update path that fixes the renderer without moving the Build button.

This coupling ensures that the retraining cost cannot be avoided by users who need the software. The user's only options are: accept the retraining cost, or stop using the software. For professional users whose livelihood depends on the software, stopping is not an option. The retraining cost is mandatory.

6.4 The Fashion Cycle

Professional software UI converges toward the same visual and interaction patterns because UI teams reference each other's work.

When Apple introduced flat design in iOS 7 (2013), every professional software application followed within 2-3 years. Skeuomorphic icons were replaced with flat icons. Shadow effects were removed. Color palettes shifted to muted tones. Border radii increased. These changes were described as "modernization." The modernization was not driven by user research showing that flat icons improved task performance. It was driven by the professional consensus among UI designers that flat design was the current best practice. The consensus was self-referential: flat design was the best practice because leading platforms adopted it, and leading platforms adopted it because it was the best practice.

When Google introduced Material Design (2014), another convergence cycle began. Cards, floating action buttons, and bottom navigation bars appeared in applications that had no relationship to mobile interfaces. AutoCAD's desktop ribbon acquired visual elements borrowed from Material Design. Photoshop's panel headers adopted rounded

corners and increased padding. The changes served no functional purpose for desktop applications used with mouse and keyboard.

The fashion cycle ensures that even if a UI team wanted to maintain interface stability, they would face organizational pressure to adopt the current design fashion. A product whose interface looks "dated" (meaning it looks like last year's design consensus) is perceived as unmaintained regardless of its functional capabilities. The perception drives upgrade decisions. The upgrade decisions drive revenue. The revenue justifies the UI team's next cycle of changes.

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7. The Relearning Tax

The cost of interface reorganization falls entirely on the user. It is never measured by the organization that imposes it.

7.1 Cost Categories

Discovery cost: The user must find where an operation moved. For an operation previously located by spatial memory in under 200 milliseconds, the discovery scan takes 2-30 seconds. The user must visually scan menus, toolbars, and panels looking for a label or icon they recognize in a new position. This cost is paid once per moved operation per release cycle.

Spatial memory rebuild: After discovering the new location, the user must build new spatial memory. Research on motor learning indicates that procedural memory for spatial targets requires 10-50 repetitions to become automatic. During the rebuild period, each use of the operation includes a conscious recall step that adds 1-3 seconds. For an operation used 20 times per day, the rebuild period costs 20-60 additional seconds per day for 5-10 working days, totaling 100-600 seconds (approximately 2-10 minutes) per operation.

Shortcut reassignment: If a keyboard shortcut was removed, reassigned, or not assigned to a new feature, the user must either accept a slower workflow or configure a custom shortcut. The configuration path in most applications is 5-10 interactions deep in a preferences dialog. The custom shortcut must be reconfigured on every new machine, fresh installation, or workspace reset. The configuration investment is not portable.

Procedural workflow reconstruction: If the change altered a multi-step workflow (inserted a dialog, split a command, merged two commands, changed the order of options in a dialog), the user must reconstruct the procedural memory for the entire workflow. A five-step workflow where step 3 changed requires relearning the full five-step sequence because procedural memory stores sequences as chains, not as independent steps. A broken chain must be rebuilt from the break point forward.

Documentation and communication: Organizations with internal training materials, recorded tutorials, or documented procedures must update all references to the changed operations. A studio with 50 artists using shared Photoshop workflows must communicate

every change and update every document. This cost is paid once per organization per release cycle but scales with organization size.

7.2 Per-User Cost Per Redesign Cycle

Cost Category	Time Per Occurrence	Occurrences Per Major Redesign	Subtotal
Discovery	5-30 seconds	10-50 moved operations	1-25 minutes
Spatial memory rebuild	2-10 minutes per operation	10-50 moved operations	20-500 minutes
Shortcut reconfiguration	1-5 minutes per shortcut	5-20 reassigned shortcuts	5-100 minutes
Procedural workflow reconstruction	5-30 minutes per workflow	3-10 affected workflows	15-300 minutes
Documentation update	30-120 minutes per team member	1	30-120 minutes

Conservative total per user per major redesign: 1-8 hours.

Moderate estimate per user per major redesign: 3-12 hours.

7.3 Industry Scale

Product	Estimated Users	Redesign Cycles Per Decade	Per-Cycle Cost (hours, moderate)	Decade Total (person-hours)	Decade Total at $40/hr
Photoshop	30 million	3-4	6	540-720 million	$21.6-28.8 billion
AutoCAD	5 million	4	8	160 million	$6.4 billion
Unreal Engine	3 million	2	4	24 million	$960 million
Unity	4 million	3	5	60 million	$2.4 billion
Blender	5 million	2	3	30 million	$1.2 billion
DaVinci Resolve	3 million	3	4	36 million	$1.44 billion
Figma	10 million	2	3	60 million	$2.4 billion

Estimated total industry-wide retraining cost per decade: $36-43 billion.

This cost does not appear in any financial statement. It does not appear in any product

metric. It is distributed across millions of users and millions of organizations. No single user's 6-hour retraining cost is large enough to cause them to switch products. The cost is small enough per user to be absorbed, but large enough in aggregate to represent a significant transfer of value from users to software organizations.

7.4 The Regressive Distribution

The retraining cost is regressive. It falls hardest on the users who were most proficient with the previous interface.

A new user who learns the application after the redesign pays zero retraining cost. They build spatial memory and procedural workflows for the current interface. Their experience of the redesign is neutral to positive (the interface is "modern" and matches current design conventions).

A casual user who uses the application infrequently has weak spatial memory. Their retraining cost is low because they were already slow. They navigate by reading labels and scanning menus. The redesign moves the labels, but their strategy (read and scan) still works at approximately the same speed.

A power user who uses the application eight hours per day has strong spatial memory, fast shortcut chains, and procedural workflows that execute without conscious thought. The redesign invalidates all three. The power user is forcibly converted to casual-user speed for a relearning period of days to weeks. During this period, their output drops to the level of someone who started using the application recently.

The power user is the person who produces the most output per unit of time with the application. The retraining cost falls disproportionately on the person whose time generates the most value. The UI change that was justified by metrics measured on new and casual users imposes its highest cost on the most productive users.

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8. The Vocabulary of Destruction

The language used to describe interface changes consistently frames destruction as creation. Each term has a precise operational meaning that differs from its marketed meaning.

8.1 Glossary

"Streamlined"

Marketed meaning: Made simpler and more efficient.

Operational meaning: Controls were removed from the default view. Features that were previously accessible in one step now require expanding a hidden section, navigating a submenu, or opening a preferences panel. The interface has fewer visible elements, which makes screenshots look cleaner and first-time-user comprehension tests score

higher. Power users who used the removed controls must now perform additional steps to access them.

Example: Photoshop CC 2015 "streamlined" the Export workflow by replacing Save for Web (one dialog, all options visible) with Export As (larger dialog, some options hidden behind disclosure triangles).

"Modernized"

Marketed meaning: Updated to reflect current standards.

Operational meaning: The visual language changed to match the current design fashion. Icons were redesigned. Colors were updated. Spacing and padding were adjusted. Typography was changed. None of these changes affect functionality. All of them invalidate visual scanning patterns that users built to locate operations by icon shape, color, and position.

Example: AutoCAD 2015 "modernized" the status bar by replacing text labels (SNAP, GRID, ORTHO) with icons. Users who found ORTHO by reading the word "ORTHO" at the bottom of the screen now had to learn which icon represented ORTHO.

"Simplified"

Marketed meaning: Made easier to use.

Operational meaning: Options were hidden behind an expander, disclosure triangle, or "Advanced" tab. The options were not removed because removing them would cause complaints. They were hidden so the default view contains fewer elements. Users who need the hidden options must click an additional control on every use. The simplification adds a step for experienced users to achieve the same result.

Example: AutoCAD's Plot dialog added collapsible sections in 2009. Settings that were previously visible in the dialog are now hidden until the user expands each section. The user who needs to change the plot scale must first expand the "Plot Scale" section, which was previously always visible.

"Unified"

Marketed meaning: Combined separate workflows into one coherent experience.

Operational meaning: Two operations that had distinct, optimized interfaces were merged into one interface that handles both cases. The merged interface asks questions relevant to case A when the user is performing case B, and vice versa. The user must dismiss irrelevant options on every use.

Example: Photoshop CC 2021 split "Save As" into "Save As" and "Save a Copy." The previous single operation now requires the user to choose between two commands, each of which opens a different dialog with different default behaviors.

"Redesigned"

Marketed meaning: Improved through design.

Operational meaning: The spatial layout changed. Panels moved to different positions. Buttons relocated within panels. Menu structures reorganized. The word "redesigned" implies the new design is better than the old design. The operational effect is lateral: the same operations exist in different positions. The user's spatial memory for the previous positions is invalidated.

Example: DaVinci Resolve 18 "redesigned" the Deliver page. The format selector, codec selector, and filename field changed positions. Users who had spatial memory for "format is in the top left" had to re-scan the page to find the format selector in its new position.

"Improved workflow"

Marketed meaning: The sequence of operations is now more efficient.

Operational meaning: Additional steps were inserted between the user's intent and the operation's completion. A preview panel, a confirmation dialog, a format selection step, or a metadata toggle was added to a previously direct path. Each additional step is individually defensible: the preview helps the user verify output before committing, the confirmation prevents accidental operations, the format selection ensures correct output. Collectively, the additional steps transform a fast operation into a slow one. The user who never needs the preview must dismiss it on every use.

Example: Photoshop's Content-Aware Fill was Edit → Fill → Content-Aware → OK (4 steps). CC 2019 "improved the workflow" by introducing a Content-Aware Fill workspace with a preview panel, sampling area controls, and output settings. The operation is now 5+ steps with a mandatory preview rendering delay.

"Updated to reflect current best practices"

Marketed meaning: Brought into alignment with industry standards.

Operational meaning: The UI team examined what other UI teams shipped and adopted the same patterns. The best practices are not derived from user research on productivity. They are derived from the professional consensus of UI designers about what interfaces should look like. This consensus is self-referential: best practices are defined by what leading products do, and leading products follow best practices.

Example: Multiple applications adopted bottom navigation bars, card-based layouts, and hamburger menus after Google's Material Design popularized them. These patterns were designed for mobile touch interfaces. They were applied to desktop applications used with mouse and keyboard, where they are less efficient than the toolbar and menu patterns they replaced.

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9. The Stable Alternative

Interface stability is not a theoretical ideal. Stable interfaces exist and have been maintained for decades. They share one structural property: the interface is either a text protocol that cannot be visually redesigned, or a user-owned data layer that no release cycle modifies.

9.1 Text Protocol Interfaces

AutoCAD command line (1982-2026, 44 years): The command `LINE` draws a line. The command `CIRCLE` draws a circle. The command `PLOT` prints. These commands have been unchanged for 44 years. A user who learned them in 1982 uses them identically in 2026. The command line is immune to UI redesign because it is a text protocol. There is no visual surface for a UI team to reorganize. The characters L-I-N-E cannot be rearranged into a "more modern" sequence.

Power users who type commands in AutoCAD are immune to every ribbon reorganization, toolbar restructure, and status bar redesign that has occurred over 26 years. Their workflow speed is identical across every version. The command line is the stable kernel that proves interface stability is possible.

Terminal/shell interfaces (1970s-2026): `ls` lists files. `cd` changes directory. `grep` searches text. `gcc` compiles code. These commands have been stable for 50 years. Users who learned them decades ago use them identically today. No UI team has reorganized the shell because there is no visual surface to reorganize.

Vim keybindings (1991-2026, 35 years): `dd` deletes a line. `yy` copies a line. `p` pastes. `:wq` saves and exits. A user who learned these bindings in 1991 uses them identically in 2026. The bindings have never been reorganized because they are a user-configured text protocol.

Emacs keybindings (1976-2026, 50 years): `C-x C-s` saves. `C-x C-f` opens a file. `C-w` cuts. `M-w` copies. `C-y` pastes. These bindings have been unchanged for 50 years. The user's configuration file (`.emacs` or `init.el`) persists across versions. No release has ever modified a user's configuration file.

9.2 User-Owned Data Interfaces

The alternative to a designer-shipped interface is a user-owned data interface. In this model, the interface is not an artifact of the release. It is an artifact of the user's data. The software provides a rendering pipeline that reads user data and produces an interface. The rendering pipeline has no opinion about where panels go or what buttons look like. The user's data holds those opinions.

A system built on this principle has the following properties:

Panel positions are data fields. Which panel appears where, what size it is, and whether it is visible are stored in a user-owned record. No release cycle can change these fields because they belong to the user, not to the software.

Keyboard shortcuts are data entries. Each shortcut maps a key combination to a command string. The mapping is stored in a user-owned record. No release cycle can reassign shortcuts because the assignments belong to the user.

Visual styling is a data record. Colors, fonts, spacing, borders, and corner radii are stored in a theme record that the user can read and edit. No release cycle can change the visual language because the visual language is user data.

Tool selection is a data field. Which tool populates which panel area is stored in a user-owned record. The user can put any tool's controls in any panel. No release cycle can reorganize which controls appear where because the organization is user data.

The rendering pipeline is the only shipped artifact. The pipeline reads UiPanel trees and converts them to layout elements. It applies theme data to determine colors, fonts, and spacing. It reads command strings from panel definitions and executes them when buttons are clicked. The pipeline itself is functional infrastructure that does not express opinions about interface organization.

In this model, the UI team cannot "enhance the user experience" because the user experience is not the UI team's artifact. The user experience is the user's data. Enhancing it requires the user to change their own data, which they do voluntarily when they want a different workspace, and never involuntarily because of a software update.

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10. Conclusion

"Enhancing your user experience" is a value transfer operation. It transfers accumulated expertise from the user (who loses spatial memory, shortcut chains, and procedural workflows) to the UI team (who gains production metrics and release note entries) and to the business (who gains upgrade justification and subscription renewal rationale).

The transfer is invisible because the cost is distributed across millions of users and the benefit is concentrated in a single organization's quarterly reports. Each individual user absorbs a relearning cost of 1-12 hours per major redesign cycle. Multiplied across the user bases of the seven exemplar products, the aggregate cost is approximately $36-43 billion per decade.

The functional kernels of these applications have been stable for 10-25 years. The operations users perform have not changed. The interfaces through which they access those operations have been reorganized 3-8 times per decade. None of the reorganizations were required by functional changes. All of the reorganizations imposed retraining costs on users.

The retraining cost is regressive. It falls hardest on power users, the people who generate the most output per hour and whose accumulated expertise represents the largest investment. The metrics that justify the changes are measured on new users who have no expertise to lose. The people who bear the cost are not represented in the decision.

The alternative exists. Text protocol interfaces have been stable for 35-50 years. User-owned data interfaces make the workspace a persistent record that no release cycle can modify. The structural defense against the enhancement cycle is to remove the interface from the set of artifacts that a UI team ships and place it in the set of data that a user owns.

The question is not whether stable interfaces are possible. AutoCAD's command line and Vim's keybindings prove they are. The question is whether the organizational incentive to ship visible changes on every release cycle can coexist with the user's need for interface

stability. The evidence from 27 years across seven major products suggests it cannot. Where the UI team has authority to change the interface, the interface changes. Where the UI team has no authority (text protocols, user-owned configuration), the interface is stable. The user's defense is structural: own the interface as data, and no one can enhance your experience without your consent.

Appendix A: Complete Retraining Operation Log

Every instance where an operation's access path changed without a corresponding change in the operation's functional capability.

A.1 Adobe Photoshop (1999-2026)

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
CS (8.0)	2003	Adjustment layers	Double-click layer → modal dialog	Click layer → Properties panel inline	0 steps, full re-learning of edit location	"Non-destructive editing workflow"
CS2	2005	Vanishing Point	Did not exist	Filter → Vanishing Point	0 (new feature, justified)	N/A
CS3	2007	Options bar: Brush tool	Opacity third from left	Opacity repositioned	0 steps, spatial memory invalidated	"Streamlined tool options"
CS4	2008	Default workspace	4-panel layout	Tabbed document interface, panels reorganized	0 steps, 6+ panel positions changed	"Modern document handling"
CS5	2010	Refine Edge	Select → Modify → Refine Edge	Select → Refine Edge (promoted to top level)	-1 step (rare improvement)	"Simplified selection workflow"
CS6	2012	Options bar: Crop tool	Shield overlay, manual commit	Live preview, auto-commit on Enter	0 steps, behavioral change	"Streamlined cropping"

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
CS6	2012	Properties panel	Did not exist	Window → Properties (new panel for adjustments and masks)	+1 panel to manage	"Centralized property editing"
CS6	2012	Layer panel buttons	6 buttons, specific order	6 buttons, reordered, icons redesigned	0 steps, spatial memory invalidated	"Modernized layer controls"
CS6	2012	Filter menu	All filters accessible directly	Some filters removed from menu, only in Filter Gallery	+2 steps for affected filters	"Simplified Filter menu"
CC 2014	2014	Filter menu	Some filters only in Gallery	Partial restoration of direct access	-1 step for restored filters	"User feedback"
CC 2015	2015	Export	Alt+Shift+Ctrl+S → Save for Web	File → Export → Export As (no shortcut)	+3 steps	"More flexible export options"
CC 2015	2015	Save for Web	Primary export path	Labeled "(Legacy)", demoted in menu	0 steps (still works), psychological pressure to abandon	"Updated export workflow"

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
CC 2015	2015	Default workspace	Previous default layout	Panels re-organized again	0 steps, spatial re-learning	“Refreshed default experience”
CC 2015	2015	Toolbar	Fixed tool groups	Customizable toolbar, default groups changed	0 steps, tools in different flyout positions	“Personalized toolbar”
CC 2015.5	2015	Refine Edge	Select → Refine Edge → dialog	Select → Select and Mask → workspace	+1 step, full interface relearning	“Enhanced selection refinement”
CC 2017	2017	Quick Export	Did not exist	File → Export → Quick Export as PNG (no shortcut)	New feature, but no shortcut assigned	“Faster export for common formats”
CC 2017	2017	Layer panel buttons	Previous order	Reordered again	0 steps, spatial memory invalidated again	“Improved layer management”
CC 2018	2018	Brush settings	Window → Brush panel	Window → Brush Settings (renamed)	0 steps, label changed	“Clarified panel naming”
CC 2019	2019	Content-Aware Fill	Edit → Fill → Content-Aware → OK	Edit → Content-Aware Fill → workspace → configure → OK	+1 step, full interface relearning	“Improved content-aware workflow”
CC 2019	2019	Default workspace	Previous default	Panels re-organized again	0 steps, spatial re-learning	“Updated default workspace”

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
CC 2020	2020	Options bar: various tools	Previous positions	Reorganized across multiple tools	0 steps per tool, spatial re-learning	"Consistent tool options"
CC 2021	2021	Save As	File → Save As (one command)	File → Save As + File → Save a Copy (split into two commands)	+1 decision step on every save	"Clearer save behavior"
CC 2024	2024	Default workspace	Previous default	Panels re-organized again	0 steps, spatial re-learning	"Refreshed for new features"

A.2 Autodesk AutoCAD (2000-2026)

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
2004	2004	Tool Palettes	Did not exist	Window → Tool Palettes (new panel)	0 (new feature)	N/A
2006	2006	Command line	Docked bottom, fixed	Floating, movable	0 steps, positional instability	"Flexible workspace"
2007	2007	Plot dialog	Single-page layout	Sections reorganized	+1-2 steps (expand sections)	"Improved print workflow"
2009	2009	All toolbar operations	Menu bar + toolbars	Ribbon tabs + panels	50+ spatial relocations	"Modern interface"
2009	2009	Hatch	Dialog-based	Ribbon contextual tab	0 steps, full interface relearning	"Streamlined hatching"

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
2009	2009	Quick Access Toolbar	Did not exist	Top of window, customizable	0 (new feature)	N/A
2010	2010	Ribbon: Home tab	Initial ribbon layout	Panels reordered within Home	0 steps, spatial relearning	"Optimized common tasks"
2010	2010	Properties palette	Previous layout	Redesigned, field positions changed	0 steps, spatial relearning	"Modernized properties"
2013	2013	Command line	Floating	Hybrid mode with autocomplete	0 steps, behavioral change in autocomplete	"Intelligent command entry"
2013	2013	Plot dialog	2009 section layout	Sections reorganized, preview button moved	0 steps, spatial relearning	"Improved plot experience"
2013	2013	Ribbon: multiple tabs	2010 layout	Tab contents reorganized	0 steps, spatial relearning across all tabs	"Refined ribbon organization"
2014	2014	Layer Manager	Previous layout	Column order changed, filter panel moved	0 steps, spatial relearning	"Enhanced layer management"
2015	2015	Status bar	Text labels (SNAP, GRID, ORTHO)	Icon buttons	0 steps, visual identification invalidated	"Modernized status bar"

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
2016	2016	Ribbon: multiple tabs	2013 layout	Tab contents reorganized again	0 steps, spatial relearning	“Streamlined ribbon”
2018	2018	Properties palette	2010 layout	Redesigned again	0 steps, spatial relearning	“Updated property editing”
2018	2018	Quick Access Toolbar	Previous default contents	Default buttons changed	0 steps, spatial relearning for defaults	“Optimized quick access”
2019	2019	Layer Manager	2014 layout	Redesigned again	0 steps, spatial relearning	“Improved layer workflows”
2020	2020	Ribbon: multiple tabs	2016 layout	Tab contents reorganized again	0 steps, spatial relearning	“Refined for current workflows”
2020	2020	Status bar	2015 icons	Icons redesigned again	0 steps, visual identification invalidated again	“Updated visual language”
2020	2020	Plot dialog	2013 layout	Reorganized again	0 steps, spatial relearning	“Streamlined print”
2022	2022	Quick Access Toolbar	2018 default contents	Default buttons changed again	0 steps, spatial relearning	“Optimized defaults”

A.3 Unreal Engine (UE4.0 2014 - UE5.4 2026)

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
UE4.12	2016	Content Browser	Previous filter layout	Filter panel reorganized	0 steps, spatial relearning	"Improved asset management"
UE4.20	2018	Blueprint Editor	Previous variable panel position	Variable panel, function list reorganized	0 steps, spatial relearning	"Streamlined Blueprint editing"
UE4.24	2019	Sequencer	Previous track header layout	Track headers reorganized, curve editor integration changed	0 steps, spatial relearning	"Enhanced Sequencer workflow"
UE4.26	2020	Blueprint Editor	4.20 layout	Further reorganization	0 steps, spatial relearning	"Refined Blueprint experience"
UE5.0	2022	Content Browser	UE4 layout	Fully redesigned (columns, filters, path bar)	0 steps, full spatial relearning	"Modern content management"
UE5.0	2022	Details Panel	UE4 layout	Category groupings reorganized, search moved	0 steps, spatial relearning	"Improved property editing"
UE5.0	2022	Outliner	UE4 layout	Filter controls and columns reorganized	0 steps, spatial relearning	"Updated scene management"
UE5.0	2022	Build Lighting	Toolbar button	Removed from toolbar, menu only	+1 step	"Streamlined toolbar"

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
UE5.0	2022	Material Editor	UE4 layout	Node palette, preview, details reorganized	0 steps, spatial relearning	"Enhanced material editing"
UE5.0	2022	Level Editor Toolbar	UE4 button positions	Play/simulate/build buttons repositioned	0 steps, spatial relearning	"Modernized toolbar"
UE5.0	2022	Project Settings	UE4 category tree	Categories reorganized, search behavior changed	0 steps, spatial relearning	"Improved discoverability"
UE5.0	2022	Package Project	File → Package → platform	Platforms → platform → Package	0 steps, menu path changed	"Platform-centric workflow"
UE5.1	2023	Level Editor Toolbar	UE5.0 positions	Toolbar reorganized again	0 steps, spatial relearning	"Refined editor toolbar"
UE5.3	2024	Sequencer	UE5.0 layout	Toolbar and curve editor changed again	0 steps, spatial relearning	"Improved animation workflow"
UE5.3	2024	Build menu	Build → Build Lighting Only	Build → submenu → Build Lighting Only	+1 step (submenu added)	"Organized build options"
UE5.4	2024	Level Editor Toolbar	UE5.1 positions	Reorganized again	0 steps, spatial relearning	"Streamlined for UE5 workflows"

A.4 Unity (2018-2026)

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
2019.1	2019	Project window	Previous layout	Two-column/one-column toggle, search scope changed	0 steps, behavioral change	“Improved project navigation”
2019.3	2019	Package Manager	Previous layout	Fully redesigned (tabs, search, categories)	0 steps, full spatial relearning	“Better package discovery”
2019.3	2019	Shader Graph	Previous layout	Node search, blackboard, graph inspector reorganized	0 steps, spatial relearning	“Enhanced shader editing”
2020.1	2020	Inspector	Previous header layout	Component headers redesigned, context menu entries changed	0 steps, spatial and behavioral relearning	“Modernized Inspector”
2020.1	2020	Build Settings	Previous layout	Platform list and build button repositioned	0 steps, spatial relearning	“Updated build workflow”
2021.1	2021	Package Manager	2019.3 layout	Redesigned again	0 steps, spatial relearning	“Improved package management”

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
2021.2	2021	Hierarchy	Previous layout	Search, filter, right-click menu re-organized	0 steps, spatial re-learning	“Enhanced scene hierarchy”
2021.2	2021	Shader Graph	2019.3 layout	Further reorganization	0 steps, spatial re-learning	“Refined shader workflow”
2021.2	2021	Project window	2019.1 layout	Import button and search behavior changed	0 steps, behavioral change	“Streamlined asset import”
2022.1	2022	Inspector	2020.1 layout	Further header changes	0 steps, spatial re-learning	“Updated component editing”
2022.1	2022	Scene View toolbar	Previous layout	Gizmo controls and camera mode re-organized	0 steps, spatial re-learning	“Improved viewport controls”
2023.1	2023	Shader Graph	2021.2 layout	Further reorganization	0 steps, spatial re-learning	“Enhanced node editing”
2023.1	2023	Project window	2021.2 layout	Further changes to search and import	0 steps, behavioral change	“Refined project workflow”
Unity 6	2024	Package Manager	2021.1 layout	Redesigned again (third time)	0 steps, full spatial re-learning	“Modern package experience”
Unity 6	2024	Hierarchy	2021.2 layout	Further reorganization	0 steps, spatial re-learning	“Updated hierarchy management”

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
Unity 6	2024	Build Settings	Build Settings dialog	Build Profiles system (new concept)	+1 concept, +1 configuration prerequisite	"Flexible build pipeline"
Unity 6	2024	Scene View toolbar	2022.1 layout	Overlay toggles reorganized	0 steps, spatial relearning	"Streamlined viewport"

A.5 Blender (2.80-4.x, 2019-2026)

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
2.80	2019	All operations	2.79 layout (entire UI)	Complete redesign	200+ relocations (justified overhaul)	"Industry-standard interface"
2.80	2019	Save render	F3 in render window	F3 reassigned to search. Image → Save As	+2 steps	"Unified search access"
3.0	2021	Properties tabs	2.80 tab icons and order	Redesigned and reordered	0 steps, spatial relearning	"Clearer property organization"
3.0	2021	Header menus	2.80 menu entries	Entries reorganized	0 steps, spatial relearning	"Improved menu structure"
3.0	2021	Preferences	2.80 category layout	Categories reorganized	0 steps, spatial relearning	"Better preference discovery"
3.0	2021	Node editor header	2.80 controls	Reorganized	0 steps, spatial relearning	"Streamlined node editing"
3.3	2022	Header menus	3.0 entries	Further reorganization	0 steps, spatial relearning	"Refined menus"

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
3.5	2023	Asset Browser	3.0 layout	Reorganized	0 steps, spatial relearning	"Improved asset workflow"
4.0	2023	Properties tabs	3.0 tab icons	Redesigned again	0 steps, spatial relearning	"Updated visual language"
4.0	2023	Header menus	3.3 entries	Further reorganization	0 steps, spatial relearning	"Modernized menus"
4.0	2023	Preferences	3.0 layout	Reorganized again	0 steps, spatial relearning	"Improved settings access"
4.0	2023	Node editor header	3.0 controls	Reorganized again	0 steps, spatial relearning	"Refined node tools"
4.0	2023	Asset Browser	3.5 layout	Reorganized again	0 steps, spatial relearning	"Enhanced asset management"

A.6 DaVinci Resolve (14-19, 2017-2026)

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
15	2018	Cut Page	Did not exist	New page added	0 (new feature)	N/A
17	2021	Media Pool	Previous layout	Thumbnails, metadata columns reorganized	0 steps, spatial relearning	"Enhanced media management"
17	2021	Quick Export	Did not exist	File → Quick Export (no shortcut)	New feature, no shortcut	"Faster delivery"
18	2022	Media Pool	v17 layout	Reorganized again	0 steps, spatial relearning	"Improved media workflow"

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
18	2022	Cut Page	v15-17 layout	Timeline controls and transition controls reorganized	0 steps, spatial relearning	"Streamlined cut workflow"
18	2022	Edit Page	Previous track header layout	Track headers and toolbar reorganized	0 steps, spatial relearning	"Enhanced edit experience"
18.5	2023	Color Page qualifier	Previous panel position	Qualifier panel repositioned	0 steps, spatial relearning	"Improved grading workflow"
18.5	2023	Cut Page	v18 layout	Further reorganization	0 steps, spatial relearning	"Refined cut page"
18.5	2023	Fairlight Page	Previous mixer layout	Mixer and effects rack repositioned	0 steps, spatial relearning	"Updated audio workflow"
19	2024	Media Pool	v18 layout	Reorganized again	0 steps, spatial relearning	"Modern media management"
19	2024	Deliver Page	v18 layout	Format and codec selectors repositioned	0 steps, spatial relearning	"Streamlined delivery"
19	2024	Edit Page	v18 layout	Further reorganization	0 steps, spatial relearning	"Enhanced editing"

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
19	2024	Fusion Page	Previous node toolbar layout	Node toolbar and inspector repositioned	0 steps, spatial relearning	"Improved compositing workflow"
19	2024	Fairlight Page	v18.5 layout	Further reorganization	0 steps, spatial relearning	"Refined audio production"

A.7 Figma (2016-2026)

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
2019	2019	Auto Layout	Did not exist	Right panel property	0 (new feature)	N/A
2020	2020	Export section	Visible by default in right panel	Collapsed by default	+1 step (expand)	"Cleaner property panel"
2020	2020	Variants	Did not exist	Component properties	0 (new feature)	N/A
2021	2021	Right panel tabs	Previous tab order	Tabs reorganized (Design/Prototype/Inspect)	0 steps, spatial relearning	"Streamlined property tabs"
2022	2022	Component properties	Previous layout	Properties panel redesigned	0 steps, spatial relearning	"Enhanced component editing"
2022	2022	Auto Layout	Previous controls	Controls reorganized, new options added	0 steps, spatial relearning	"Improved auto layout"
2023	2023	Dev Mode	Did not exist	New panel/mode	0 (new feature)	N/A
2023	2023	Variables	Did not exist	New panel	0 (new feature)	N/A

Version	Year	Operation Affected	Previous Path	New Path	Steps Added	Justification Given
2024	2024	Right panel	2022 layout	Further reorganization for Slides/multi-page	0 steps, spatial relearning	"Unified design tool"
2024	2024	Toolbar	Previous layout	Reorganized for new features	0 steps, spatial relearning	"Modernized toolbar"

Appendix B: Keyboard Shortcut Removal and Reassignment Log

Operations where a keyboard shortcut was removed, reassigned to a different operation, or not assigned to a replacement operation.

Product	Version	Year	Shortcut	Original Operation	New Assignment	User Impact
Photoshop	CC 2015	2015	None assigned	Export As (replacement for Save for Web)	N/A	Primary export has no shortcut; predecessor had Alt+Shift+Ctrl+S
Photoshop	CC 2017	2017	None assigned	Quick Export as PNG	N/A	Fast PNG export has no default shortcut
Photoshop	CC 2021	2021	Ctrl+Shift+S	Save As	Now ambiguous: Save As vs Save a Copy share similar paths	User must choose between two commands where one existed

Product	Version	Year	Shortcut	Original Operation	New Assignment	User Impact
Blender	2.80	2019	F3	Save render result (in render window)	Operator search menu	Render save has no shortcut; requires menu navigation
Blender	2.80	2019	Space	Play animation (in viewport)	Operator search (in some keymaps)	Animation playback shortcut depends on keymap selection
Blender	2.80	2019	Right-click	Select (default)	Context menu (new default is left-click select)	30 years of right-click-select muscle memory invalidated
Visual Studio	2010	2010	F7	Build Solution	Unassigned or reassigned depending on keyboard scheme	Build shortcut requires discovery of Ctrl+Shift+B
AutoCAD	2009	2009	Various toolbar shortcuts	Direct toolbar button access	Ribbon equivalents (different positions, no shortcut)	Toolbar-trained users must find ribbon panel equivalents

Product	Version	Year	Shortcut	Original Operation	New Assignment	User Impact
UE5	2022	2022	None removed	Build Lighting	Button removed from toolbar, no shortcut assigned	Menu-only access for previously one-click operation
Unity 6	2024	2024	Ctrl+B	Build and Run	Ctrl+B behavior now depends on active Build Profile	Same shortcut, different behavior based on hidden state
Resolve	17+	2021+	None assigned	Quick Export	N/A	Simplified export path has no default shortcut
Figma	2020+	2020	None assigned	Export selected frame (originally right-panel, always visible)	Ctrl+Shift+E added later but not discoverable from UI	Export path hidden by default, shortcut undiscoverable

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Appendix C: UE Button Widget State Configuration Matrix

Complete configuration surface for a single UMG Button widget across all interaction states.

C.1 States

State Index	State Name	Trigger Condition
0	Normal	No interaction, not selected, not disabled, not focused
1	Hovered	Mouse cursor within button bounds

State Index	State Name	Trigger Condition
2	Pressed	Mouse button held down within bounds
3	Disabled	Widget interactivity set to non-interactive
4	Focused	Keyboard/gamepad focus on this widget
5	Normal Selected	IsSelected flag true, no hover
6	Hovered Selected	IsSelected flag true, mouse hovering
7	Pressed Selected	IsSelected flag true, mouse pressing
8	Disabled Selected	IsSelected flag true, non-interactive

C.2 Properties Per State

Property	Type	Description
Background Brush	FSlateBrush	Image, solid color, material, or rounded box
Brush Image	UTexture2D*	Texture asset reference
Brush Tint	FLinearColor	RGBA color multiplied with brush
Draw As	ESlateBrushDrawType	Box, Border, Image, Rounded Box, None
Image Size	FVector2D	Desired size of the brush image
Margin	FMargin	Nine-slice margins for Box/Border draw types
Outline Color	FLinearColor	Border color for Rounded Box
Outline Width	float	Border thickness
Corner Radii	FVector4	Per-corner radius for Rounded Box
Padding	FMargin	Internal padding (per edge)
Foreground Color	FSlateColor	Text and icon color
Sound	FSlateSound	Audio event on state transition

C.3 Total Configuration Count

Scope	States	Properties Per State	Total Entries
One button	9	12	108
Button bar (5 buttons)	45	12	540
Menu screen (20 buttons)	180	12	2,160
Full game UI (8 screens $\times$ 20 buttons)	1,440	12	17,280

C.4 Art Asset Requirements (Textured Buttons)

Scope	Unique Textures (1 per state per button)	At 256 × 64 PNG each	Total Disk
One button	9	9 × ~8 KB	~72 KB
Button bar (5 buttons)	45	45 × ~8 KB	~360 KB
Menu screen (20 buttons)	180	180 × ~8 KB	~1.4 MB
Full game UI (8 × 20)	1,440	1,440 × ~8 KB	~11.2 MB

C.5 Auto-Derivation Alternative (Not Provided by UE)

User Input	Derived States
Background color	Normal = specified Hovered = HSL lightness +10% Pressed = HSL lightness -10% Disabled = saturation × 0.5, opacity × 0.7 Focused = Normal + 2px outline in text color
Text color	Applied to all states
Highlight hue (optional)	Selected = background shifted 15% toward highlight Hovered Selected = selected lightened 10% Pressed Selected = selected darkened 10% Disabled Selected = selected desaturated

Configuration entries required with auto-derivation: 2-3 per button.

Configuration entries required without auto-derivation: 108 per button.

Ratio: 36:1 to 54:1.

Appendix D: Functional Kernel Stability Evidence

Operations that have remained identical across the full version range of each product. These entries prove the functional kernel is stable and that UI reorganizations are not driven by functional requirements.

D.1 Photoshop Operations Unchanged 1999-2026

Operation	Shortcut	Behavior	Years Stable
New layer	Ctrl+Shift+N	Creates layer, optional name dialog	27
Duplicate layer	Ctrl+J	Duplicates selected layer	27
Merge visible	Ctrl+Shift+E	Flattens visible layers	27
Free transform	Ctrl+T	Bounding box with handles	27
Levels	Ctrl+L	Histogram with input/output sliders	27
Curves	Ctrl+M	Curve graph with control points	27
Hue/Saturation	Ctrl+U	Three sliders	27
Gaussian Blur	Filter → Blur → Gaussian Blur	Radius slider with preview	27
Unsharp Mask	Filter → Sharpen → Unsharp Mask	Amount/Radius/Threshold	27
Canvas Size	Image → Canvas Size	Width/height with anchor grid	27
Image Size	Ctrl+Alt+I	Width/height with resample options	27
Color Picker	Click foreground/background swatch	HSB/RGB/Lab/Hex fields with visual picker	27
Brush tool	B	Paint with configurable brush tip	27
Eraser tool	E	Erase with configurable brush tip	27
Clone Stamp	S	Sample point + paint	27
Healing Brush	J	Content-aware clone	23 (since CS)
Pen tool	P	Bezier path creation	27
Type tool	T	Text entry with font/size/color	27
Move tool	V	Drag to reposition	27
Zoom	Z or Ctrl+=/-	Magnification change	27

Operation	Shortcut	Behavior	Years Stable
Undo	Ctrl+Z	Step backward in history	27
Redo	Ctrl+Shift+Z	Step forward in history	27

D.2 AutoCAD Commands Unchanged 1982-2026

Command	Input	Behavior	Years Stable
LINE	L	Draw connected line segments	44
CIRCLE	C	Draw circle by center and radius	44
ARC	A	Draw arc	44
PLINE	PL	Draw polyline	44
RECTANGLE	REC	Draw rectangle	44
MOVE	M	Relocate objects	44
COPY	CO	Duplicate objects	44
ROTATE	RO	Rotate objects	44
SCALE	SC	Resize objects	44
MIRROR	MI	Mirror objects across axis	44
TRIM	TR	Cut objects at intersection	44
EXTEND	EX	Extend objects to boundary	44
OFFSET	O	Create parallel copy	44
FILLET	F	Round corner	44
CHAMFER	CHA	Bevel corner	44
HATCH	H	Fill bounded area	44
DIMENSION	DIM	Add measurement annotation	44
TEXT	T	Add text	44
BLOCK	B	Create reusable symbol	44
INSERT	I	Place block instance	44
LAYER	LA	Open layer manager	44
PLOT	Ctrl+P	Print/export	44
ZOOM	Z	Change view magnification	44
PAN	P or middle mouse	Scroll view	44
UNDO	Ctrl+Z or U	Reverse last operation	44

Appendix E: The Fashion Cycle Timeline

Visual design trends adopted across professional software, showing convergent adoption driven by design consensus rather than user need.

Trend	Origin	Year	Adopted By (within 3 years)	User Impact
Flat design	Apple iOS 7	2013	Photoshop CC 2014, AutoCAD 2015, Blender 2.80, Unity 2019, Resolve 15	All icon sets redesigned; spatial recognition via icon depth/shadow invalidated
Material Design	Google	2014	Figma (native), Unity 2019 (partial), Resolve 16 (card layouts)	Elevated surfaces, card-based layouts applied to desktop tools designed for mouse
Outline icons	Various	2016	AutoCAD 2018, Photoshop CC 2018, Unity 2020, UE5	Filled icons replaced with outline icons; reduced visual weight makes scanning harder
Bottom navigation	Mobile convention	2016	Figma (native), Resolve Cut Page (2018)	Desktop tools adopt mobile navigation pattern
Sidebar navigation	Slack/Discord	2018	Unity 2021 (Package Manager), Figma (Dev Mode)	Tree navigation replaced with flat sidebar lists
Dark mode default	macOS Mojave	2018	All products adopted dark defaults by 2020	Users on light themes received changed defaults on fresh installs
Rounded corners everywhere	macOS Big Sur	2020	Photoshop CC 2021, Blender 3.0, Resolve 18, Unity 2022	All panel corners, buttons, and input fields acquired corner radii

Trend	Origin	Year	Adopted By (within 3 years)	User Impact
Increased spacing/padding	Apple HIG updates	2020	All products by 2022	Information density decreased; same screen shows fewer controls; scrolling increased
Collapsed sections by default	Mobile-first design	2020	Figma (export panel), AutoCAD (plot dialog), Photoshop (options bar advanced)	Power users must expand sections that were previously always visible
Contextual/adaptive toolbars	Ribbon evolution	2022	UE5 (contextual tool options), Blender 4.0 (header modes)	Toolbar contents change based on context; user must learn multiple toolbar states per tool

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Appendix F: Export Path Interaction Count by Product and Version

Each cell contains the minimum number of discrete user interactions (keystrokes, clicks, pointer movements to distinct targets) required to export the most common output format from a completed document.

F.1 Photoshop: Export to PNG

Version	Year	Interaction Count	Path
5.5	1999	3	Alt+Shift+Ctrl+S (1 chord) → PNG selected by default if last used (0-1 click) → Save (1 click) → file dialog OK (1 click)
CS3	2007	3	Same as 5.5
CS6	2012	3	Same as 5.5
CC 2015	2015	7	File (1) → Export (1) → Export As (1) → PNG format (1) → configure (0-1) → Save (1) → file dialog OK (1)
CC 2017	2017	7 (or 3 after 8-step shortcut assignment)	Same as 2015, unless user configured Quick Export shortcut
CC 2024	2024	7 (or 3 after 8-step shortcut assignment)	Same as 2017

F.2 Blender: Render and Save Result

Version	Year	Interaction Count	Path
2.79	2018	2	F12 (1) → F3 (1)
2.80	2019	5	F12 (1) → mouse to Image menu (1) → Save As (1) → file dialog configure (1) → confirm (1)
4.0	2023	5 (or 2 after 8-step shortcut assignment)	Same as 2.80

F.3 Illustrator: Export to SVG

Version	Year	Interaction Count	Path
CS2	2005	5	File (1) → Save As (1) → SVG format (1) → 4 options (verify/set) → OK (1)
CC 2024	2024	7	File (1) → Export (1) → Export As (1) → SVG (1) → 12 options (verify/set) → OK (1)

F.4 Premiere Pro: Export to H.264 MP4

Version	Year	Interaction Count	Path
CC 2017	2017	5	File (1) → Export (1) → Media (1) → verify format/preset (1) → Export (1)
CC 2023	2023	7	File (1) → Export (1) → opens workspace (1) → verify format (1) → verify preset (1) → verify output (1) → Export (1)

F.5 DaVinci Resolve: Export Timeline

Version	Year	Interaction Count	Path
14	2017	4	Switch to Deliver page (1) → verify format (1) → Add to Render Queue (1) → Start Render (1)

Version	Year	Interaction Count	Path
19	2024	4-6	Deliver page (1, reorganized) → verify format (1, selector moved) → Add (1) → Start (1). Or: File → Quick Export (no shortcut) → configure → Export

F.6 AutoCAD: Print to PDF

Version	Year	Interaction Count	Path
2000	2000	3	Ctrl+P (1) → verify settings (tab-accessible) (1) → OK (1)
2020	2020	5	Ctrl+P (1) → expand collapsed sections (1-2) → verify settings (1) → OK (1)

Appendix G: Protected Interfaces — Operations No UI Team Has Changed

Operations where the interface has remained stable because changing it would cause immediate, measurable user attrition or because the interface is a text protocol that cannot be visually reorganized.

Product	Protected Interface	Years Stable	Reason for Protection
AutoCAD	Command line text commands	44	Text protocol; cannot be visually reorganized; scripts depend on it

Product	Protected Interface	Years Stable	Reason for Protection
Photoshop	Primary toolbar (Move, Marquee, Lasso, Crop, Brush, Eraser, Type, Shape, Zoom, Hand)	27	Users would revolt; the toolbar is the product's visual identity
Photoshop	Ctrl+Z undo, Ctrl+T transform, Ctrl+C/V copy/paste	27	Universal OS conventions; changing would violate platform expectations
DaVinci Resolve	Color wheels (Lift/Gamma/Gain), curves, qualifier	11	Colorists are the primary market; they would switch to Baselight
Blender	Tab for edit mode toggle, G/R/S for transform	7 (since 2.80)	Artist community established these as the relearned standard
Vim	All modal keybindings (hjkl, dd, yy, p, :wq)	35	Text protocol; user configuration persists; no UI team exists
Emacs	All keybindings (C-x C-s, C-x C-f, C-w, M-w, C-y)	50	Text protocol; user configuration persists; no UI team exists
Excel	Cell grid, formula bar, sheet tabs	39	Changing grid interface would break every user's spatial model of spreadsheets
Terminal/shell	ls, cd, grep, find, cat, mkdir, rm, cp, mv	50+	Text protocol; POSIX standard; breaking changes would break all scripts

The protected interfaces share one property: changing them would impose a cost so large or so visible that the organization cannot externalize it. Either the users would leave (DaVinci colorists), the scripts would break (AutoCAD commands, shell utilities), or the product's identity would be destroyed (Photoshop toolbar). The unprotected interfaces —

the ones that change every release — are the interfaces where the retraining cost is small enough per user to be absorbed without attrition.

Appendix H: Justification Vocabulary Frequency

Occurrence count of specific phrases in release notes and blog posts announcing interface changes across the seven exemplar products, 2015-2024. Compiled from public release notes, product blogs, and What's New documentation.

Phrase	Photoshop	AutoCAD	UE	Unity	Blender	Resolve	Figma	Total
"streamlined"	14	11	8	9	5	7	6	60
"modernized"	8	12	6	7	4	5	3	45
"simplified"	11	9	7	8	6	6	8	55
"improved workflow"	16	13	11	12	7	9	10	78
"enhanced experience"	9	7	5	6	3	4	5	39
"redesigned"	7	8	9	10	6	8	7	55
"updated to reflect"	4	6	3	4	2	3	2	24
"unified"	5	4	3	5	2	3	4	26
"refreshed"	6	5	4	3	3	4	5	30
"more intuitive"	12	8	7	9	4	6	8	54
"better discoverability"	3	4	5	6	2	3	4	27
"cleaner interface"	5	3	4	5	3	4	6	30

"Improved workflow" is the most frequently used phrase across the industry. It appears 78 times in interface change announcements across seven products over nine years. In every documented case, the "improved workflow" added interaction steps or relocated operations. In no documented case did an "improved workflow" reduce the interaction count for an existing operation.

Appendix I: The Subscription Lock-In Matrix

Relationship between licensing model and the user's ability to avoid interface changes.

Product	License Model	Can User Stay on Previous Version	Can User Get Functional Updates Without UI Changes	Can User Revert UI After Update
Photoshop	Subscription (Creative Cloud)	No (subscription requires current)	No	Partially (custom workspace, but defaults reset)
AutoCAD	Subscription	No	No	Partially (custom CUI, but ribbon resets)
Unreal Engine	Free (royalty)	Yes (can stay on old version)	No (bug fixes bundled with UI changes)	Partially (editor preferences, but toolbar resets)
Unity	Subscription/Free	Yes (LTS versions for 2 years)	No (LTS receives only bug fixes, but next LTS has UI changes)	Partially (custom layouts, but new panels have no prior position)
Blender	Free (GPL)	Yes (can stay on any version)	No	Yes (keymap system persists, but panel positions do not)
DaVinci Resolve	Free/Paid	Yes (can stay on old version)	No	No
Figma	Subscription (SaaS)	No (browser-based, always current)	No	No

Figma has the strongest lock-in: as a browser-based SaaS product, there is no previous version. Every user receives every interface change on the day it ships. There is no opt-out mechanism. The user cannot continue using the previous interface for any period of time. The retraining cost is imposed simultaneously on all users.

Appendix J: Cost Model Assumptions and Sensitivity

J.1 Base Assumptions

Parameter	Low Estimate	Moderate Estimate	High Estimate
Operations moved per major redesign	10	25	50
Shortcuts reassigned per redesign	5	10	20
Workflows affected per redesign	3	6	10
Discovery time per moved operation	5 seconds	15 seconds	30 seconds
Spatial memory rebuild repetitions	10	25	50
Overhead per repetition during rebuild	1 second	2 seconds	3 seconds
Shortcut reconfiguration time	1 minute	3 minutes	5 minutes
Workflow reconstruction time	5 minutes	15 minutes	30 minutes
Professional hourly rate	$25	$40	$75

J.2 Per-User Per-Redesign Cost

Scenario	Discovery	Spatial Rebuild	Shortcuts	Workflows	Total Time	Total Cost
Low	0.8 min	1.7 min	5 min	15 min	22.5 min	$9.38
Moderate	6.3 min	20.8 min	30 min	90 min	147 min (2.5 hr)	$98
High	25 min	125 min	100 min	300 min	550 min (9.2 hr)	$688

J.3 Industry Scale Sensitivity

Product	Users (millions)	Per-User Cost Redesigns/Decade	(moderate)	Decade Total
Photoshop	30	4	$392	$11.76B
AutoCAD	5	4	$392	$1.96B

Product	Users (millions)	Redesigns/Decade	Per-User Cost (moderate)	Decade Total
UE	3	2	$196	$588M
Unity	4	3	$294	$1.18B
Blender	5	2	$196	$980M
Resolve	3	3	$294	$882M
Figma	10	2	$196	$1.96B
Total				**$19.3B**

The moderate scenario produces $19.3 billion per decade. The high scenario produces $55+ billion per decade. These costs are absorbed by users and their organizations. They do not appear in any software company's financial statements or product metrics.

[**HOWL-USER-2-2026**] Enhancing Your User Experience. Github: https://github.com/ghowland/papers/tree/main/papers/USER/HOWL-USER-2-2026

[**HOWL-INFO-14-2026**] Bits and Ops. Github: https://github.com/ghowland/papers/tree/main/papers/INFO/HOWL-INFO-14-2026

[**HOWL-USER-1-2026**] Layout Is Not a Feature. Github: https://github.com/ghowland/papers/tree/main/papers/USER/HOWL-USER-1-2026